

Agent Lifecycle Management: Network Management Agent Observability, Intervention and Control

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IETF ICON Side Meeting

Wednesday 22 July 2026

8:00 - 9:30 (Vienna Time)

Park Suite 4

Background and Recap

- AINETOPS github was created (<https://github.com/IETF-OPS-AD/AINETOPS>) to
 - Coordinate or provide classification for AI related work distributed in different WGs
 - Keep track of BOF, Side meetings, Mailing lists, Hackathon Projects
- AI for Network Operation wiki page (<https://wiki.ietf.org/e/en/group/ops/aiops>) provides guideline for AI driven Network Operation work
- AI topics can be break down into Network for AI and AI for Network
 - Network for AI such as: AI Network Multicast, AIDC, FANN
 - AI for Network such as: DAWN, AgenticOPS side meeting
- AI Topics are not new
 - AIPref WG
 - Vocabulary for expressing AI-related preferences
 - Attaching preferences with content in IETF-defined protocols and formats
 - Webbotauth WG
 - Cryptographically authenticating method for AI Crawlers
 - NMOP WG
 - Network Anomaly architecture, Lifecycle and Metadata Semantics
 - Network Incident YANG
 - NMRG
 - Research challenges in Coupling AI and Network Management
 - AI Deployment Consideration
 - Agentic AI Architecture and Problem Statement
 - ...

Why are we here?

- We always believe AI for good
 - UC Cases: service complaint Handling, IP+Optical cross layer cross domain fault management, etc.
 - Value: OPEX/CAPEX reduction, ROI service (Service Manager), site visit reduction (Field Engineer), etc.
- What if AI for bad/AI Agent goes wrong?
 - **Non-Deterministic Execution Paths**
 - same input with different chains of tool calls, retries, and reasoning steps
 - **Cascading Failures That Look Like Success**
 - An agent can return a confident-sounding wrong answer after silently failing to retrieve context
 - **Hidden Cost Accumulation**
 - single user query might trigger 15 LLM calls across different models
 - agent loop can burn through a monthly token budget in hours
 - **Extreme Latency Variance**
 - Same query might complete in 800ms or 30 seconds depending on the reasoning path

Why are We here

- What can we do about it if Agent goes wrong
 - Can We See it? -> Agent Observability
 - Can we set boundary to prevent agent from violating policies? Agent Control
 - Can we stop high risk operation? -> Human Oversight
- Goal of this side meeting:
 - Follow up the previous AgenticOPS side meeting
 - <https://github.com/danielkinguk/AgenticOps>
 - Explore key areas where standardization is required to support the design, implementation, and operation of Network Management Agent Observability, Intervention and Control

Meeting Agenda

- Opening Remarks(IETF meeting rules, agenda arrangement, and objective)
- 5 min Moderators
- Network Management Agent Observability, Intervention and Control (ICON)
Problem Statement -10 min + 5min Qin Wu
 - Reference: <https://datatracker.ietf.org/doc/draft-wnd-opsawg-icon-ps/>
- Network Management Agent Observability, Intervention and Control (ICON)
Architecture and Requirements 10 min +5 min Qiufang Ma
 - Reference: <https://datatracker.ietf.org/doc/draft-mcw-opsawg-icon-requirements/>
- Current work on Observability, Traceability at AAIF 10min + 5min Daniel King
- Open Discussion on this new proposed Initiative 40 min All
- Wrap-up / Next steps 5 min Moderators

Discussion

- Is the problem Real? What else is missing?
- Is IETF right place to develop solution to address this problem?
- Who are interested to contribute?